



PROJECT IN MATHEMATICS

On Permutation Groups

by

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Abstract

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Chapter 1

Introduction

The motivation behind the research whose the results can be found largely within this work, was the desire to find an answer to a deceptively simple question: “What is the process by which methods for solving problems are found?”

The question arose within me in relation to a long fascination with the problem-solving process employed when finding methods for solving “twisty puzzles” – such as the classic Rubik’s cube. It was this initial focus which led me to study *permutation groups*, and initially formulate my thoughts and findings in terms thereof.

...

Chapter 2

Preliminaries

2.1 Groups

Definition 1 (Group). In mathematics, a *group* G denotes a set of elements $\{a, b, c, \dots\}$ and an accompanying binary operator $*$, together satisfying the following conditions:

1. *Closure*: For every ordered pair of elements in the group, the binary product thereof (as defined by the binary operator) is also an element of that group; $\forall a, b \in G \ (a * b \in G)$.
2. *Associativity*: The order wherein ordered pairs of elements are evaluated has no consequence on the resulting product; $\forall a, b, c \in G \ ((a * b) * c = a * (b * c))$.
3. *Existence of Identity*: There exists in the group an *identity* element, the omission of which from any expression containing any other element or elements has no consequence on the evaluation of that expression; $\exists e \in G \ \forall a \in G \ (e * a = a * e = a)$.
4. *Existence of Inverse*: Each element in the group has its own *inverse*, who's product therewith evalutes to the identity; $\forall a \in G \ \exists b \in G \ (a * b = e)$, where e is the identity.

Definition 2 (Commutativity). Let G be a group. Then it is *commutative* (or *Abelian*) if $\forall a, b \in G \ (a * b = b * a)$.

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ together with *addition* constitutes an Abelian group with 0 acting as its identity (since $\forall a \in \mathbb{Z} \ (0 + a = a + 0 = a)$). The negative numbers exist as the inverses of the positives, and vice versa. We know addition is associative, and since $\mathbb{Z}$ is infinite in each direction, we will never find a pair of elements therein whose sum is not also an element therein. This group can also be expressed as additively *generated* by the element 1, since 1 and its inverse -1 together with $+$ can express the whole group. This generative definition is denoted $G = (\langle\{1\}\rangle, +) = (\mathbb{Z}, +)$. The group generated by 2 – to give another example – is the group of all *even* integers.

Definition 3 (Cyclicity). A group which can be generated by a single element is called a *cyclic* group.

When the operator is given implicitly, a cyclic group with the generator g can be written as $\langle g \rangle$.

Definition 4 (Subgroup). Let G be a group under a binary operator $*$. Then $H \subset G$ is a *subgroup* of G if H is also a group under $*$.

2.2 Permutations

Definition 5 (Permutation). A *permutation* π of a set A is a bijective map $\pi : A \rightarrow A$.

We write π^n for a permutation π to denote $\pi \circ \dots \circ \pi$ where π is composed with itself $n \in \mathbb{N}$ times. π^{-n} denotes the composition $\pi^{-1} \circ \dots \circ \pi^{-1}$ where the inverse π^{-1} is composed with itself $n \in \mathbb{N}$ times. The permutation π^0 always equals the identity.

Definition 6 (Permutation Group). A *permutation group* on a set A is a group of permutations of A under composition.

Definition 7 (Support). The *support* $\underline{\pi}$ of a permutation π of A is the set $\{a \in A \mid \pi(a) \neq a\}$.

A permutation whose support is the empty set equals the identity. We say that a permutation π of A *fixes* an element $a \in A$ if $a \notin \underline{\pi}$.

Definition 8 (Cycle). A *cycle* is a permutation π such that $\forall a, b \in \underline{\pi} \exists \phi \in \langle \pi \rangle (\phi(a) = b)$.

Since the support of the identity is the empty set, it follows that the identity also a cycle.

Definition 9 (Subcycle of an Element). The *subcycle* $\overset{a}{\pi}$ of an element $a \in A$ within a permutation π on A is cycle on A whose support $\underline{\overset{a}{\pi}}$ is the set $\{\phi(a) \mid \phi \in \langle \pi \rangle\}$ and where $\forall b \in \underline{\overset{a}{\pi}} \left(\overset{a}{\pi}(b) = \pi(b) \right)$.

Definition 10 (Subcycle of a Permutation). A *subcycle* $\phi \in \pi$ of a permutation π on A is a cycle on A where $\forall a \in \underline{\phi} (\phi(a) = \pi(a))$.

Note that if π is a cycle, then π is its own subcycle.

Definition 11 (Subpermutation of a Permutation). A *subpermutation* $\phi \subset \pi$ of a permutation π of A is a permutation $\phi \neq \pi$ on A where $\forall a \in \underline{\phi} (\phi(a) = \pi(a))$.

Definition 12 (Orbit). The *orbit* of an element $a \in A$ in a permutation π on A is the support of the subcycle of a in π .

Definition 13 (Transposition). A *transposition* is a cycle π with $|\underline{\pi}| = 2$.

Theorem 1. *Let π be a permutation of a set A . Then there exists a family of cycles $(\pi_i)_{i \in I}$ on A with pairwise disjoint supports such that for each $a \in \underline{\pi}$, there exists unique $i \in I$ where $\pi(a) = \pi_i(a)$.*

Proof. Let $a_{k+1} \in \underline{\pi} \setminus \bigcup_{i \in \{1, \dots, k\}} \underline{\pi_i}$ where $k \in \mathbb{N}$ and each π_i is a cycle satisfying $\forall n \in \mathbb{Z} (\pi_i^n(a_{k+1}) = \pi^n(a_{k+1}))$. Then

□

In *cycle notation*, Theorem 1 is used to express permutations in a convenient way. Here, each cycle in the expressed permutation is written as a parenthesized list of elements, where each element therein is mapped to the succeeding element save for the last one, which is mapped to the first. For example, the permutation $(a, b, c)(d, e)$ maps $a \mapsto b$, $b \mapsto c$ and $c \mapsto a$, as well as $d \mapsto e$ and $e \mapsto d$. Note that this permutation is the same as the permutation $(c, a, b)(e, d)$, but not the same as $(a, c, b)(d, e)$.

In this text, permutations containing symbols or numbers with multiple digits will be written with dividing commas, whilst permutations containing exclusively the numbers 0 through 9 will be written therewithout – for example $(1234) = (1, 2, 3, 4)$.

Lemma 1. *Let α and β be permutations of A . Then α and β do not commute if and only if there is an element $c \in \underline{\alpha} \cap \underline{\beta}$ where the cycles of α and β wherein c is contained are neither equal nor each other's inverses.*

Proof. Let α and β be permutations of A and $\underline{\alpha} \cap \underline{\beta} = \emptyset$. Then $\forall b \in A (\alpha(b) = b \vee \beta(b) = b)$. Thus for each $b \in A$ both $(\alpha \circ \beta)(b)$ and $(\beta \circ \alpha)(b)$ evaluate to the same expression, which will be either $\alpha(b)$, $\beta(b)$ or b . A permutation always commutes with itself, so if $\alpha_c = \beta_c$ for some $c \in \underline{\alpha} \cap \underline{\beta}$ – where α_c and β_c are the cycles of α and β wherein c occurs – then $\alpha_c \circ \beta_c = \beta_c \circ \alpha_c$. If each cycle in α and β commute, so do they. If $\alpha_c = \beta_c^{-1}$, then the cycles neutralize each other $\alpha_c \circ \beta_c = \beta_c \circ \alpha_c = e$. Now suppose there is an element $c \in \underline{\alpha} \cap \underline{\beta}$ such that $\alpha_c \neq \beta_c$ and $\alpha_c \neq \beta_c^{-1}$. Then from $\beta(c) \notin \underline{\alpha}$ follows $(\alpha \circ \beta)(c) = \beta(c)$ and thereof $(\beta \circ \alpha)(c) \neq \beta(c)$. If $\beta(c) \in \underline{\alpha}$, then $(\alpha \circ \beta)(c) \neq \beta(c)$

□

2.2.1 On set notation for permutations

Since (by Theorem 1) permutations can be represented by sets of cycles, we can use set notation for permutations. For example, $\alpha \in \beta$ implies that α is a cycle ...

Chapter 3

The Conjugates and Commutators of Permutations

Herein, all lowercase geek letters are implied to be elements in a permutation group G on A . Elements of A are represented by lowercase latin characters. Further, the operator $\circ$ will henceforth be implicit, and thus omitted from expressions. As an example, take $\pi \circ \phi = \pi\phi$.

Definition 14 (Conjugate). The *conjugate* $\searrow_{\beta}^{\alpha}$ of β by α is the composition $\alpha^{-1}\beta\alpha$.

Definition 15 (Commutator). The *commutator* $\times_{\beta}^{\alpha}$ of α and β is the composition $\beta^{-1}\alpha^{-1}\beta\alpha$.

These two constructions – though they can seem arbitrary at first – have interesting and useful properties when working within groups – specifically when the supports of their two permutations share elements.

In the trivial case where they share no element, the permutations commute, so the conjugate of α and β is β , and the commutator thereof is the identity; $\alpha \cap \beta = \emptyset$ implies that $\alpha^{-1}\beta\alpha = \beta$ and $\beta^{-1}\alpha^{-1}\beta\alpha = e$.

Lemma 2. Let $\underline{\alpha} \cap \underline{\beta} = \{c\}$. Then $\searrow_{\beta}^{\alpha} = (\alpha^{-1}(c), \beta(c), \dots, \beta^{-1}(c))$.

Proof. For each $b \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\}$, $\alpha(b) \notin \underline{\beta}$. Since $\beta\alpha(b) = \alpha(b)$, the conjugate can be rewritten for b : $(\alpha^{-1}\beta\alpha)(b) = (\alpha^{-1}\alpha)(b) = b$. In the case of $\alpha^{-1}(c)$, $(\alpha^{-1}\beta\alpha\alpha^{-1})(c) = (\alpha^{-1}\beta)(c) = \beta(c)$, since $\beta(c) \notin \underline{\alpha} \implies (\alpha^{-1}\beta)(c) = \beta(c)$. For each $d \in \underline{\beta} \setminus \{\beta^{-1}(c), c\}$, $d, \beta(d) \notin \underline{\alpha}$, so $(\alpha^{-1}\beta\alpha)(d) = \beta(d)$. Lastly, $\beta^{-1}(c)$ maps to $(\alpha^{-1}\beta\alpha\beta^{-1})(c) = (\alpha^{-1}\beta\beta^{-1})(c) = \alpha^{-1}(c)$. This gives the function

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.1)$$

whose subcycle of c is written as $(\alpha^{-1}(c), \beta(c), \dots, \beta^{-1}(c))$. □

Corollary 1. If β is a transposition, then so is $\searrow_{\beta}^{\alpha}$.

Proof. If β is a transposition, then $\beta = (c, \beta(c))$ and $\beta(c) = \beta^{-1}(c)$. Using (3.1) we get

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \emptyset \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.2)$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta(c) \end{cases} \quad (3.3)$$

$$= (\alpha^{-1}(c), \beta(c)). \quad (3.4)$$

□

Lemma 3. Let $\underline{\alpha} \cap \underline{\beta} = \{c\}$. Then $\searrow_{\beta}^{\alpha} = (c, \beta^{-1}(c), \alpha^{-1}(c))$.

Proof. Using (3.1) from Lemma 2, we get

$$\searrow_{\beta}^{\alpha}(x) = (\beta^{-1} \searrow_{\beta}^{\alpha})(x) = \begin{cases} (\beta^{-1})(x) & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ (\beta^{-1}\beta)(c) & : x = \alpha^{-1}(c) \\ (\beta^{-1}\beta)(x) & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ (\beta^{-1}\alpha^{-1})(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.5)$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c), c\} \\ \beta^{-1}(x) & : x = c \\ c & : x = \alpha^{-1}(c) \\ x & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.6)$$

$$= \begin{cases} x & : x \in (\underline{\alpha} \cup \underline{\beta}) \setminus \{\alpha^{-1}(c), \beta^{-1}(c), c\} \\ c & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \\ \beta^{-1}(c) & : x = c \end{cases} \quad (3.7)$$

$$= (c, \beta^{-1}(c), \alpha^{-1}(c)). \quad (3.8)$$

□

Lemma 4. Let $\underline{\alpha} \cap \underline{\beta} = \{c_1, \dots, c_n\}$, $\alpha \neq \beta$ for $n \geq 2$ where $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ for $k \in \{1, \dots, n\}$. Then $\searrow_{\beta}^{\alpha} = (\alpha^{-1}(c_1), c_1, \dots, c_{n-1}, \beta(c_n), \dots, \beta^{-1}(c_1))$.

Proof. Let $a \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\}$. Then $\alpha(a) \notin \beta$, so $(\alpha^{-1}\beta\alpha)(a) = (\alpha^{-1}\alpha)(a) = a$. Conversely, each $b \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\}$ maps to $(\alpha^{-1}\beta\alpha)(b) = \beta(b)$, since $b, \beta(b) \notin \underline{\alpha}$. As for the elements $\{c_1, \dots, c_{n-2}\}$, none thereof diverge from the common path $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ for $k \in \{1, \dots, n\}$, wherefore $c_j \xrightarrow{\alpha} c_{j+1} \xrightarrow{\beta} c_{j+2} \xrightarrow{\alpha^{-1}} c_{j+1}$ in $\searrow_{\beta}^{\alpha}$ for each $j \in \{1, \dots, n-2\}$. Lastly, $\alpha^{-1}(c_1)$ maps to $(\alpha^{-1}\beta\alpha\alpha^{-1})(c_1) = (\alpha^{-1}\beta)(c_1) =$

$\alpha^{-1}(c_2) = c_1$, $\beta^{-1}(c_1)$ maps to $(\alpha^{-1}\beta\alpha\beta^{-1})(c_1) = (\alpha^{-1}\beta\beta^{-1})(c_1) = \alpha^{-1}(c_1)$ and c_{n-1} maps to $(\alpha^{-1}\beta\alpha)(c_{n-1}) = (\alpha^{-1}\beta)(c_n) = \beta(c_n)$. This gives the function

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ c_1 & : x = \alpha^{-1}(c_1) \\ \beta(x) & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ \beta(c_n) & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.9)$$

$$\implies \searrow_{\beta}^{\alpha} = \left(\alpha^{-1}(c_1), c_1, \dots, c_{n-1}, \beta(c_n), \dots, \beta^{-1}(c_1) \right). \quad (3.10)$$

□

Lemma 5. Let α and β be permutations with the same properties as those in Lemma 4. Then $\times_{\beta}^{\alpha} = (\alpha^{-1}(c_1), \beta^{-1}(c_1))(c_{n-1}, c_n)$.

Proof. From (3.9) from Lemma 4 we get

$$\times_{\beta}^{\alpha}(x) = \begin{cases} \beta^{-1}(x) & : x \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ (\beta^{-1}\beta)(x) & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ (\beta^{-1}\beta)(c_n) & : x = c_{n-1} \\ (\beta^{-1}\alpha^{-1})(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.11)$$

$$= \begin{cases} x & : x \in \{\alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ \beta^{-1}(c_n) & : x = c_n \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ x & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ c_n & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.12)$$

$$= \begin{cases} x & : x \in \{c_{n-1}, c_n, \alpha^{-1}(c_1), \beta^{-1}(c_1)\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \\ c_n & : x = c_{n-1} \\ c_{n-1} & : x = c_n \end{cases} \quad (3.13)$$

$$= (\alpha^{-1}(c_1), \beta^{-1}(c_1))(c_{n-1}, c_n). \quad (3.14)$$

□

Lemma 6. Let

DEFINE PERMUTATION SHAPE.

Lemma 7. Let $\emptyset \neq \underline{\alpha} \cap \underline{\beta} \neq \underline{\alpha} \cup \underline{\beta}$ and $\forall \beta' \in \beta \left(\underline{\beta}' \subseteq \underline{\alpha} \vee \underline{\beta}' \cap \underline{\alpha} = \emptyset \right)$. Now, let γ be a permutation $\gamma \subset \beta$ such that $\underline{\beta} \setminus \underline{\gamma} \cap \underline{\alpha} = \emptyset$ and $\underline{\gamma} \subseteq \underline{\alpha}$.

Proof. Let $\beta = \beta_1, \dots, \beta_n \beta'_1, \dots, \beta'_m$ and $\beta' = \beta'_1, \dots, \beta'_m$. Then

$$\bowtie_{\beta}^{\alpha} = \left\{ \begin{array}{ll} x & : x \in \underline{\beta} \setminus \underline{\beta'} \\ \beta'(x) & : x \end{array} \right. .$$

□

Chapter 4

The 2×3 Game

The “ 2×3 game” is a simple puzzle with six tiles in a grid of numbers and a set of allowed moves which permute them. The goal of the game is to sort the grid into a “solved” state from some scrambled state.

$$\begin{bmatrix} 5 & 4 & 2 \\ 3 & 6 & 1 \end{bmatrix} \xrightarrow{\text{Move}_1} \dots \xrightarrow{\text{Move}_n} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Formally, the game can and will herein henceforth be modelled as follows. Let $G := (\langle \mathfrak{M} \rangle, \circ)$ be a group compositionally generated by the set of moves $\mathfrak{M} := \{(123), (456), (14)\}$, named $U = (123)$ for *upper row*, $L = (456)$ for *lower row* and $S = (14)$ for *side flip*. A *problem* is a permutation $p \in G$, and a *solution* thereto is a vector $\mathbf{s} \in \mathfrak{M}^*$ such that

$$\bigcirc_{i=n}^1 \mathbf{s}_i = \mathbf{s}_n \circ \mathbf{s}_{n-1} \circ \dots \circ \mathbf{s}_1 = p$$

where n denotes the length of $\mathbf{s}$. Note that $\mathbf{s}$ may include the implicitly given inverses of the group generators. Hereafter, $\bigcirc_{i=n}^1 \mathbf{s}_i$ will be written as $\acute{\mathbf{s}}$.

The size of the group $|G|$ is 720, which shows that $G = S_6$, since $|S_6| = 720$. This fact can also be proven in the following way. Each transposition between the upper and lower row can be found with the nested conjugate

$$\begin{aligned} T_1(x, y) &= \left(L^{-y+3+1} \right)^{-1} \left(\left(U^{-x+1} \right)^{-1} (S) \left(U^{-x+1} \right) \right) \left(L^{-y+3+1} \right) \\ &= L^{y-4} U^{x-1} S U^{-x+1} L^{-y+4} \\ &= L^{y-1} U^{x-1} S U^{-x+1} L^{-y+1}, \end{aligned}$$

where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$. As an example, let $p = (26)$. Then

$$\begin{aligned}
T_1(2, 6) &= L^{6-1}U^{2-1}SU^{-2+1}L^{-6+1} \\
&= L^5U^1SU^{-1}L^{-5} \\
&= (456)^5(123)^1(14)(123)^{-1}(456)^{-5} \\
&= (465)(123)(14)(132)(456) \\
&= (465)(24)(456) \\
&= (2, 6).
\end{aligned}$$

This gives – after some reductions – the solution $s = (L, U^{-1}, S, U, L^{-1})$ visualized below.

$$\begin{array}{ccccc}
\boxed{\begin{matrix} 1 & 6 & 3 \\ 4 & 5 & 2 \end{matrix}} & \xrightarrow{L} & \boxed{\begin{matrix} 1 & 6 & 3 \\ 2 & 4 & 5 \end{matrix}} & \xrightarrow{U^{-1}} & \boxed{\begin{matrix} 6 & 3 & 1 \\ 2 & 4 & 5 \end{matrix}} \\
\downarrow S & & & & \\
\boxed{\begin{matrix} 2 & 3 & 1 \\ 6 & 4 & 5 \end{matrix}} & \xrightarrow{U} & \boxed{\begin{matrix} 1 & 2 & 3 \\ 6 & 4 & 5 \end{matrix}} & \xrightarrow{L^{-1}} & \boxed{\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{matrix}}
\end{array}$$

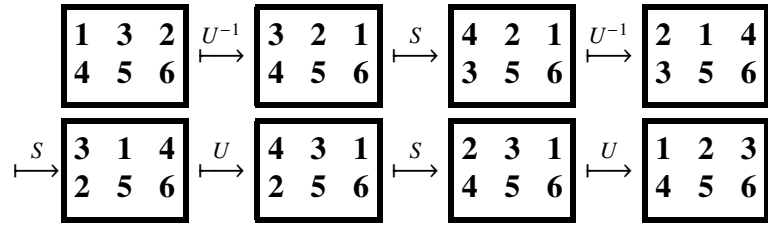
Each transposition between elements on the same row can be found with the nested conjugate

$$\begin{aligned}
T_2(x, y, R) &= \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right)^{-1} \left((R^{-y+1})^{-1}(S)(R^{-y+1}) \right) \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right) \\
&= (R^{x-1}SR^{-x+1})^{-1}(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{-(-x+1)}S^{-1}R^{-(x-1)})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{x-1}SR^{-x+1})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= R^{x-1}SR^{-x+1}R^{y-1}SR^{-y+1}R^{x-1}SR^{-x+1} \\
&= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1}
\end{aligned}$$

where $x, y \in R$ and $R \in \mathfrak{M} \setminus \{S\}$. As an example, let $p = (23)$. Since 2 and 3 belong to the upper row, R is set to U :

$$\begin{aligned}
T_2(2, 3, U) &= U^{2-1}SU^{-2+3}SU^{2-3}SU^{-2+1} \\
&= U^1SU^1SU^{-1}SU^{-1} \\
&= (123)(14)(123)(14)(123)^{-1}(14)(123)^{-1} \\
&= (123)(14)(123)(14)(132)(14)(132) \\
&= (123)(14)(24)(14)(132) \\
&= (123)(12)(132) \\
&= (23).
\end{aligned}$$

This gives the solution $\mathbf{s} = (U^{-1}, S, U^{-1}, S, U, S, U)$ visualized below.



Since a transposition of any two elements herein can be composed using the generators in $\mathfrak{M}$ – in other words, since the group G includes every transposition of the elements 1 through 6 – it follows that the group must include every permutation on the six elements, and thereby equal S_6 .

Chapter 5

New

Since G is the symmetric group S_6 , there must exist a map $T_1(x, y) = (x, y)$, where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$.

From Corrolary 1, we know that since $\zeta_1(x, y)$ is a transposition, then $\zeta_1 = \searrow_{\beta_1}^{\alpha_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$. Similarly, from $|\underline{\beta_1}| = 2$ follows that $\beta_1 = \searrow_{\beta_2}^{\alpha_2}$ where $\beta_2 = (c_1, c_2)$ and $\alpha_2(x) = c_2$. Now, from $\mathfrak{M}$ we can find permutations matching these properties

$$T_1(x, y) = \searrow_{\substack{U^{-x+1} \\ S}}^{L^{-y+3+1}}$$

$$\begin{aligned} T_2(x, y, R) &= \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right)^{-1} \left((R^{-y+1})^{-1}(S)(R^{-y+1}) \right) \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right) \\ &= R^{x-1} S R^{-x+y} S R^{x-y} S R^{-x+1} \end{aligned}$$

Since G is the symmetric group S_6 , there must be a map $\zeta : \{1, \dots, 6\}^2 \rightarrow \mathcal{P}(\mathfrak{M}^{<\omega})$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $s \in \zeta(x, y) \implies \acute{s} = (x, y)$. In other words: there must exist a set of solutions to each transposition in S_6 .

Since G is the symmetric group S_6 , there must be an infinite¹ set of maps $\{1, \dots, 6\}^2 \rightarrow \mathfrak{M}^{<\omega}$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $s \in \zeta(x, y) \implies \acute{s} = (x, y)$ where ζ is such a map.

.....There is a set of solutions to T_1 , and α_1 and β_1 and α_2 and β_2 . the solution is the intersection thereof. At each level there is a unique map (for that level) to a set of solutions². It's recursive!

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The lemmas and corollaries in chapter 3 – and indeed, any other observation, general or particular (about permutations or a specific game) – together construct a formal language.

Example: π where $|\pi| = 2$ is a *pattern*, which can be substituted by another pattern: a conjugation $\searrow_{\beta_1}^{\alpha_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$ (as in the example above). The permutation $\beta_1 = (x, c_1)$ is, then, also a pattern to be substituted and so on until you have a permutation constructed from available moves ($\mathfrak{M}?$). You don't even need to evaluate them all if you know, for example, that ζ_1/T_1 exists.

All the lemmas and corollaries are in fact rewritings (with rewriting rules – in this case, formal math) ...

A *pattern* is a set p . If a set, vector, element (or other data) Ω matches the pattern p , then $\Omega \in p$. The set $\{p \mid \Omega \in p\}$ is the set of patterns to which some data Ω belongs.

Example

Let $a \gg b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned} \mathfrak{E} &:= \mathfrak{I} \cup \{e_1 \gg o \gg e_2 \mid o \in \{[“+”], [“-”]\} \wedge e_1, e_2 \in \mathfrak{E}\} \\ \mathfrak{I} &:= \mathfrak{F} \cup \{t_1 \gg o \gg t_2 \mid o \in \{[“\times”], [“\div”]\} \wedge t_1, t_2 \in \mathfrak{I}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{[(“”) \gg e \gg (“”)] \mid e \in \mathfrak{E}\} \\ \mathfrak{N} &:= \mathcal{N} \cup \mathcal{N}^* \\ \mathcal{N} &:= \{“0”, \dots, “9”\} \end{aligned}$$

$$\sigma' = \lambda_{\mathcal{I}}(\Phi, \sigma)$$

$$\begin{cases} 1 & : x < 4 \\ 0 & : x > 6 \end{cases}$$

¹It is easy to see that this set is infinite, since one can append any solution with a permutation and its inverse to get a new solution. Take the solution $s_2 = (\alpha, \beta, \gamma)$ as an example, then $\acute{s}_1 = \acute{s}_2$ where $s_2 = (\alpha, \beta, \gamma, \alpha, \alpha^{-1})$.

²These are *patterns* with rewriting rules!

Chapter 6

Going Further

Definition 16 (Grimoire). A *grimoire* Φ of a semigroup G , a set of *spells* $\mathfrak{S}_\Phi \subseteq G$ and a set of *movements* $\mathfrak{M} \subseteq \mathfrak{S}_\Phi$ is a map in $\mathfrak{S}_\Phi \rightarrow \mathcal{P}(\mathfrak{S}_\Phi^{<\omega})$ such that $s_1 * \dots * s_n = s$ – where $*$ is the binary operator belonging to G – for each $[s_1, \dots, s_n] \in \Phi(s)$ and $s \in \mathfrak{S}_\Phi \setminus \mathfrak{M}$.

Definition 17 (Exploration Function). Let a map $\Delta : [\Phi] \times G^{<\omega} \rightarrow [\Phi]$ – where $[\Phi]$ is the set of all grimoires of a subgroup G – be defined for some grimoire Φ by $\forall s \in \mathfrak{S}_\Phi \setminus \{s_1 * \dots * s_n\} \ (\Phi'(s) = \Phi(s))$ and $\Phi'(s_1 * \dots * s_n) = \Phi(s_1 * \dots * s_n) \cup \{[s_1, \dots, s_n]\}$ if $s_1 * \dots * s_n \in \mathfrak{S}_\Phi$ and $\Phi'(s_1 * \dots * s_n) = \{[s_1, \dots, s_n]\}$ if not – where $\Phi' = \Delta(\Phi, [s_1, \dots, s_n])$ and $s_1, \dots, s_n \in \mathfrak{S}_\Phi$. (Note that since $\mathfrak{S}_\Phi = \text{dom}(\Phi)$, then $\mathfrak{S}_{\Phi'} = \mathfrak{S}_\Phi \cup \{s_1 * \dots * s_n\}$ if $\Phi' = \Delta(\Phi, [s_1, \dots, s_n])$.) An *exploration function* $\mathcal{E}$ in G is a map in $[\Phi] \times [C]^{<\omega} \rightarrow [\Phi]$ defined by

$$\mathcal{E}(\Phi, [C_1, \dots, C_n]) := \bigcirc_{i=1}^n \Delta(\Phi, C_i(\mathfrak{S}_\Phi)),$$

where $[C] \subset \mathcal{P}(G) \rightarrow G^{<\omega}$ and $C(S) \in S^{<\omega}$ for each $C \in [C]$ and $S \in \mathcal{P}(G)$.

Definition 18 (Expansion Function). An *expansion function* λ for a grimoire Φ of a semigroup G is a map in $[\Phi] \times \mathfrak{S}_\Phi \times [\Pi] \rightarrow \mathfrak{M}^{<\omega}$ – where $[\Phi]$ is the set of all grimoires of G – defined by

$$\lambda(\Phi, s, \Pi) := \begin{cases} s & : \lambda_{\text{step}}(\Phi, s, \Pi) = s \\ \lambda(\Phi, \lambda_{\text{step}}(\Phi, s, \Pi), \Pi) & : \text{otherwise} \end{cases}$$

where $\lambda_{\text{step}} : [\Phi] \times \mathfrak{S}_\Phi \times [\Pi] \rightarrow \mathfrak{S}_\Phi^{<\omega}$ is defined by

$$\lambda_{\text{step}}(\Phi, s, \Pi) := \bigvee_{i=1}^{|s|} \left(\begin{cases} s_i & : s_i \in \mathfrak{M} \\ \Pi(\Phi(s_i)) & : \text{otherwise} \end{cases} \right),$$

where $[\Pi] \subset \mathcal{P}(G^{<\omega}) \rightarrow G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ holds for each $\Pi \in [\Pi]$ and $S \in \mathcal{P}(G^{<\omega})$.

Proove that grimoire-exploration with $\mathcal{E}/\Delta$ is a semigroup.

Är gruppen definierad av *likheten* mellan en mängd utforskningar/utökningar av trollformelboken?